Willis Wang

(408) 966-2909
williswang@berkeley.edu
williswang.dev
linkedin.com/in/willislwang/

Experience

Aug 2022–Present **Meta**, *Software Engineer*, Menlo Park, CA.
- Sole owner of Meta's fleet-wide metrics collection daemon (14M+ hosts, 400B+ datapoints/minute), powering observability and alerting across all Meta systems.
- Defined and executed on the daemon's multi-year technical roadmap, driving efficiency, platform support, and operational sustainability improvements.
- Led design and implementation of read request tracking system for exabyte-scale storage, building unsampled logging (7M+ QPS), data aggregation pipelines, and analytics dashboards.
- Diagnosed recurring memory pressure in metrics routers and deployed throughput-aware load balancing, eliminating a category of data loss.
- Built first end-to-end shadow environment for metrics infrastructure, enabling pre-production regression testing across 7+ distributed services.
- Drove deprecation of legacy cross-region storage proxy, saving $12.5M in infrastructure costs.
- **Technologies and Libraries:** C++, Linux, Python, Prometheus (equivalent), Kubernetes (equivalent), Chef, Thrift, Scribe

May 2021–Aug 2021 **Amazon**, *SDE Intern — AWS Internal Tooling*, Seattle, WA.
- Led design and implementation of categorization system for internal AWS account management.
- Deployed application to 150k+ end users within Amazon's Consumer Division.
- **Technologies and Libraries:** Javascript, AWS Lambda, API Gateway, DynamoDB

May 2020–Aug 2020 **8th Wall**, *Software Engineering Intern*, Mountain View, CA.
- Implemented storage, backup, retention, and restoration system for all customer code repositories.
- Built javascript tools automating internal database migrations, rollback, and CDN updates.
- **Technologies and Libraries**: Python, Javascript, AWS Lambda, S3, API Gateway, DynamoDB, Cloudwatch, Cloudfront, SNS, Elastic Beanstalk

Education

Aug 2018–May 2022 **University of California, Berkeley**, *BA Computer Science*, Berkeley, CA, 3.75.
- Graduate Operating Systems • Operating Systems & Systems Programming • Computer Security
- Graduate Parallel Computing • Programming Languages & Compilers • Database Systems
- Computer Architecture • Internet Architecture and Protocols • Data Structures

Extracurriculars

Jan 2021–May 2022 **Undergraduate Student Instructor**, *CS162 Operating Systems*, Berkeley, CA.
- Taught operating systems concepts (concurrency, virtual memory, filesystems) to a class of 400; created course content for PintOS instructional OS.

Sep 2021–May 2023 **Student Researcher**, *Fog Robotics Group*, Berkeley, CA.
- Designed scalable multicast message-passing protocol in C++ for distributed data platform.

Jan 2021–Feb 2022 **Student Researcher**, *RISE Lab*, Berkeley, CA.
- Contributed to Dataspread, a hybrid storage engine integrating spreadsheet semantics with PostgreSQL, implementing query scheduling optimizations for concurrent data access.
- Implemented out-of-order cell update scheduler maximizing cell usability and availability in Java.

Skills

Languages **Core:** C, C++, Python, Java, Scala (familiar), SQL
Sectondary: Bash, RISC-V, x86-64/ARM Assembly, OCaml, Go, Rust, Javascript

Tools/Domains Distributed Systems, Data Pipelines, Kernel Development, Software Optimization; GDB, Linux, Hadoop/Spark, Delta Lake, CUDA, MPI, OpenMP, UPC++, Numpy, LaTeX

Publications

Janurary 2022 **SCL: A Secure Concurrency Layer For Paranoid Stateful Lambdas**.
Kaiyuan Chen, Alexander Thomas, Hanming Lu, William Mullen, Jeff Ichnowski, Rahul Arya, Nivedha Krishnakumar, Ryan Teoh, **Willis Wang**, Anthony Joseph, John Kubiatowicz. (SEC 2022)

December 2021 **Delegated, Sharded, Multicast Tree for Paranoid Stateful Lambdas** .
Marcus Plutowski, **Willis Wang**, Vivek Bharadwaj (unpublished)